Vectors I Basics Worksheet: Questions

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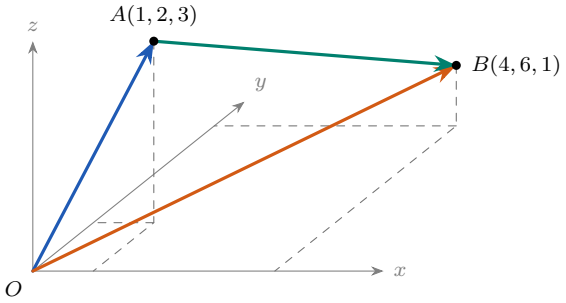
Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points and the ratio theorem. Each section has footing questions, one tricky question and a True/False check.

1 i, j, k Notation and Column Vectors

1 Write $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ as a column vector, and write $\mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$ in terms of **i**, **j** and **k**.

2 The points A and B have coordinates $(1, 2, 3)$ and $(4, 6, 1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



3 **(Tricky)** Given $\mathbf{p} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{q} = -\mathbf{i} + 3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p} - 3\mathbf{q} + \mathbf{r} = \mathbf{0}$. Give $\mathbf{r}$ in terms of **i**, **j** and **k**.

Check Your Understanding. Circle T (True) or F (False) for each statement.

	Statement	**T/F**
(a)	$3\mathbf{i} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$	T / F
(b)	The $\mathbf{j}$-component of $4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ is -1	T / F
(c)	The position vector of $P(1, -2, 0)$ is $\mathbf{i} - 2\mathbf{j}$	T / F
(d)	$\overrightarrow{AB} = \overrightarrow{OA} - \overrightarrow{OB}$	T / F
(e)	$\mathbf{i} + \mathbf{j} + \mathbf{k} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$	T / F

2 Magnitude

1 Find $|2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}|$.

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

3 **(Tricky)** Given $\mathbf{a} = \begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}| = 3$, find the possible values of t.

Check Your Understanding. Circle T (True) or F (False) for each statement.

	Statement	**T/F**
(a)	$\lvert\mathbf{i}+2\mathbf{j}-2\mathbf{k}\rvert = 1+2-2=1$	T / F
(b)	$\lvert-\mathbf{a}\rvert = \lvert\mathbf{a}\rvert$ for every vector $\mathbf{a}$	T / F
(c)	$\lvert\mathbf{a}+\mathbf{b}\rvert = \lvert\mathbf{a}\rvert + \lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$	T / F
(d)	$\lvert 3\mathbf{a}\rvert = 3\lvert\mathbf{a}\rvert$	T / F
(e)	$\left\lvert \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix} \right\rvert = -5$	T / F
(f)	$\left\lvert \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\rvert = 1$	T / F
(g)	$\lvert\mathbf{a}+\mathbf{b}\rvert \leq \lvert\mathbf{a}\rvert + \lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$	T / F

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

3 **(Tricky)** The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

Check Your Understanding. Circle T (True) or F (False) for each statement.

	Statement	**T/F**
(a)	$\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector	T / F
(b)	$\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector	T / F
(c)	$\mathbf{i}+\mathbf{j}$ is a unit vector	T / F
(d)	$\lvert\hat{\mathbf{a}}\rvert = 1$ for every non-zero vector $\mathbf{a}$	T / F
(e)	$\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel	T / F

4 Parallel Vectors ($\mathbf{a} = \lambda\mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}-6\\3\\-9\end{pmatrix}$ are parallel.

2 Find p and q such that $\begin{pmatrix}4\\p\\-2\end{pmatrix}$ is parallel to $\begin{pmatrix}-2\\3\\q\end{pmatrix}$.

3 **(Tricky)** Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix}1\\2\\-2\end{pmatrix}$.

Check Your Understanding. Circle T (True) or F (False) for each statement.

	Statement	**T/F**
(a)	If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$	T / F
(b)	$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$	T / F
(c)	$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$	T / F
(d)	$\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$	T / F
(e)	If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions	T / F

5 Collinear Points

1 Show that the points $A(1, 2, 3)$, $B(3, 5, 4)$ and $C(7, 11, 6)$ are collinear.

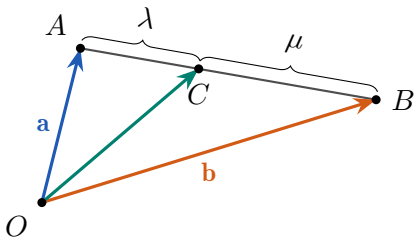
2 For the points in Question 1, find the ratio $AB : BC$.

3 **(Tricky)** The points $P(2, -1, 0)$, $Q(4, 1, 3)$ and $R(a, 7, b)$ are collinear. Find a and b.

Check Your Understanding. Circle T (True) or F (False) for each statement.

	Statement	**T/F**
(a)	If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear	T / F
(b)	If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear	T / F
(c)	$A(0,0,0)$, $B(1,1,1)$ and $C(2,2,3)$ are collinear	T / F
(d)	The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$)	T / F

6 Ratio Theorem



In the True/False table, C lies on AB with $AC : CB = \lambda : \mu$, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

1 Let $\overrightarrow{OA} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\overrightarrow{OB} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC : CB = 1 : 2$,

(b) $3AC = 2CB$,

(c) $2AB = 5CB$,

(d) Given instead that C lies on AB produced, and $3AC = 7BC$.

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3, -1, 5)$ and $(-1, 5, 1)$.

Check Your Understanding. Circle T (True) or F (False) for each statement.

	Statement	**T/F**
(a)	$\overrightarrow{OC} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$	T / F
(b)	$\overrightarrow{OC} = \dfrac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$	T / F
(c)	If $\lambda : \mu = 1 : 1$, then $\overrightarrow{OC} = \dfrac{1}{2}(\mathbf{a} + \mathbf{b})$	T / F
(d)	If $\lambda : \mu = 1 : 3$, then $\overrightarrow{OC} = \dfrac{3\mathbf{a} + \mathbf{b}}{4}$	T / F
(e)	If $\lambda : \mu = 2 : 1$, then $\overrightarrow{OC} = \dfrac{2\mathbf{a} + \mathbf{b}}{3}$	T / F

END OF WORKSHEET